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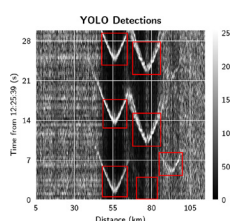
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Finding the whale song in an acoustic data haystack

Maia Chandler

Automated three-stage analysis pipeline sifts out whale calls from terabytes of distributed acoustic sensing data.



Subsea fiber optics don't only carry the internet across the ocean floor — in recent years, these sprawling cables have also been used to eavesdrop on whales.

Manually analyzing terabytes of data from hundreds of kilometers of fiber optics, however, is no small effort. To facilitate the use of this data, Truong et al. developed an automated three-stage analysis pipeline that identifies hidden fin whale calls in distributed acoustic sensing (DAS) data.

DAS sends repeated laser pulses down optical fibers and measures tiny changes in the pulse reflections to determine the location of acoustic disturbances. Fin whale calls appear as repeated hyperbolas in the resulting time-space data.

After cleaning the data to enhance signal-to-noise ratios, the team tested four signal detection methods — the Hough transform, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), template matching, and the You Only Look Once (YOLO) algorithm.

YOLO was the fastest method, taking only 2.6 seconds to process 30 seconds of data. The residual errors of candidate fin whale calls were then used in a light gradient boosting machine (LightGBM) to screen false positives.

By making it easier to detect whale calls at scale, the method could enable long-term monitoring of whale populations.

“It is a reusable project,” Truong said. “This pipeline can be leveraged to detect blue whales, humpback whales, and other marine mammals. You just need to tweak the frequency that fits what you are looking for.”

The pipeline is not limited to underwater sound detection — the team is currently adapting it to detect landslides along Norway's railroads.

“A rock falling into the track is really dangerous for passengers,” Truong said. “So, instead of detecting the hyperbolas produced by fin whales, we now detect rockfall signals.”

Source: “Automated detection of fin whales with distributed acoustic sensing in the arctic and mediterranean,” by Khanh Truong, Jo Eidsvik, Robin Andre Rørstadbotnen, Jan Petter Morten, Laurine Andres, and Anthony Sladen, *The Journal of the Acoustical Society of America* (2025). The article can be accessed at <https://doi.org/10.1121/10.0041855>.

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